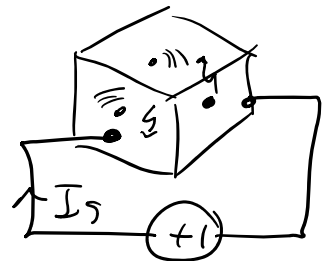


3-types of LED efficiency

- 1.) Internal Quantum Efficiency (η_{IQE}): total light inside crystal
- 2.) External Quantum Efficiency (η_{EQE}): Light exits the crystal
- 3.) Power Conversion Efficiency (η_{PCE}): input power to optical output

1) Internal Quantum Efficiency

$$\eta_{IQE} = \frac{\text{Rate of "Radiation" of Recombinations} \left(\begin{array}{c} \nearrow \text{light} \\ \text{light} \end{array} \right)}{\text{Total Recombinations (vibration of atom and Radiation)}} \\ \downarrow \begin{array}{c} \text{vibration} \\ \text{of atom} \end{array} + \nearrow \begin{array}{c} \text{light} \\ \text{radiation} \end{array} \\ = \frac{(\nearrow)}{(\text{vib}) + (\nearrow)} = \frac{\left(\frac{1}{\tau_{rad}} \right)}{\left(\frac{1}{\tau_{non-rad}} \right) + \left(\frac{1}{\tau_{rad}} \right)}$$



$$\eta_{IQE} = \frac{\text{Total internal Photons emitted per sec}}{\text{the total carriers lost per sec.}}$$

* more important for Laser Diodes.

$$\eta_{IQE} = \frac{(P_o' / hf)}{hf I_s} \quad ; \quad P_o' \text{ is the optical power generated inside material}$$

I_s is source current.

2.) External Quantum Efficiency

$$\eta_{EQE} = \frac{\text{optical Power leaving crystal}}{\text{electrons going in}}$$

Φ_e (radiant flux) Watts $\rightarrow P_o$ optical power leaving

$$\eta_{EQE} = \frac{(\Phi_e / hf)}{(I_s / q)} = \frac{q \Phi_e}{I_s hf} \quad \text{LED}$$

Note

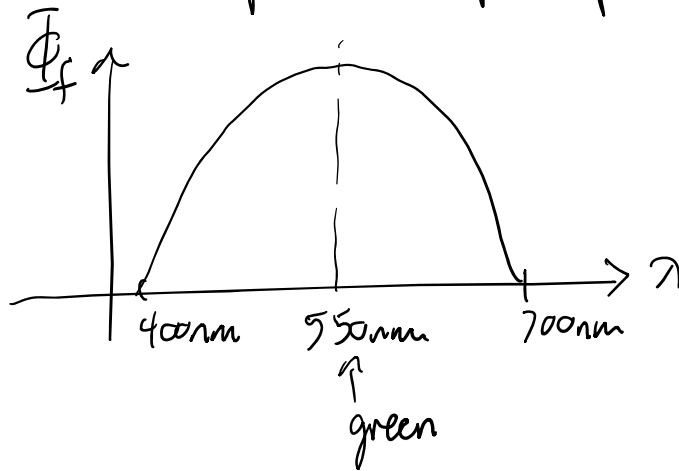
$$h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$$

$$\hbar = \frac{h}{2\pi} = 1.05457 \times 10^{-34} \text{ J} \cdot \text{s}$$

3.) Power Conversion Efficiency

$$\eta_{PCF} = \frac{P_o}{(I_s)(V_s)} \approx \eta_{EQE} \left(\frac{\tilde{E}_g}{q(V_s)} \right)$$

Luminous Flux (Φ_f): a measure of the human vision vs. optical output power.

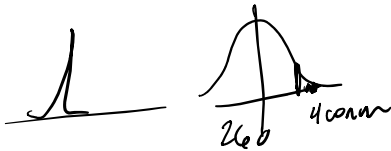


Lumens is a measure of perceived Brightness of the human eye.

Safety is important with optics

260nm $\Rightarrow$ destroys biological "things"

2W



1 lumen = 1.58 mW @ 555nm light source.

- lumen is just the power
- lux is the lumen per area.

Optical Excitation creates two types of Effects

Effect 1

